

Nicholas James Fettig

nfettig@umich.edu • (301) 919-0040 • [linkedin.com/in/nick-fettig](https://www.linkedin.com/in/nick-fettig) • fettig.dev • github.com/nickfettig

EDUCATION

University of Michigan | Ann Arbor, MI

M.S. Robotics

Aug 2026 - Dec 2027

Incoming Coursework: Math for Robotics, Robotic Systems Lab, Intro to Robot Controls

Swarthmore College | Swarthmore, PA

B.S. Engineering, B.A. Computer Science | GPA 3.86 (3.96 in-major)

May 2026

Honors & Awards: Joshua Lippincott Graduate Fellow, Tau Beta Pi, Martin S. Kapp Scholarship

TECHNICAL SKILLS

Languages: C++, C, Python, MATLAB, Bash, JavaScript/TypeScript, PostgreSQL

Robotics: ROS 2, EtherCAT (Beckhoff/TwinCAT), real-time Linux, hardware-in-the-loop (HITL) testing

Tools: CMake, Git, GDB, Valgrind, Docker, PyTorch/TensorFlow, AWS, Terraform, LaTeX

WORK EXPERIENCE

Shinkei Systems | Los Angeles, CA

May 2026 - Present

Robotics Software Engineering Intern

- Integrated load-cell force sensors, read through two Beckhoff EL3356 terminals, into an in-house motor-control service over EtherCAT; wrote SDO configuration for startup calibration and per-channel scaling
- Built a per-fish weight-extraction algorithm (ETL job) over the continuous load-cell weight stream, segmenting individual fish from the running signal to reach **<1% mean signed error at batch scale**

Reliable Robotics | Mountain View, CA

May - August 2025

Software Engineering Intern

- Led development of "Big Board," a Mission Control display used by remote pilots operating automated aircraft, integrating live ADS-B telemetry for **8 company aircraft** in a safety-critical operational loop (LitJS, AWS EC2)

Carnegie Mellon University | Pittsburgh, PA

May 2024 - May 2025

Computer Science Undergraduate Researcher

- Surveyed and classified **200+ works** on RL-based traffic signal control into a novel taxonomy of deployment challenges (sim-to-real transfer, safety constraints, infrastructure limitations)

PROJECTS

Autonomous Mobile Robot | Raspberry Pi, ROS 2, C++, ESP32

- Built a differential/mecanum-drive mobile robot from scratch: RPLIDAR A1M8 and MPU6050 IMU into an ROS 2 stack on a Raspberry Pi 5, with an ESP32-S3 handling closed-loop motor control and wheel odometry over serial

SEPTA Transit Display | Raspberry Pi, Python, HUB75

- Built a 256x128 LED matrix display (16 chained HUB75 panels) for live SEPTA train arrivals, solving panel chain addressing and coordinate mapping for the 4x4 grid; deployed as a systemd service on a Pi 5

Fish Species Classification | TensorFlow, ResNet50, Computer Vision

- Trained a ResNet50 classifier with early fusion, concatenating segmentation masks as a fourth input channel alongside RGB, achieving **97.22% classification accuracy**

Home Server | Docker, Linux, Cloudflare

- Run an Ubuntu home server hosting containerized services (personal site, blog, Jellyfin) behind key-only SSH and a Cloudflare tunnel